4.1 Rates of Change and Rectilinear Motion

Name _____ Period ____ Date _____

AP Calculus 4.1 · Student sheet

I can…

- find velocity and acceleration from a position function.
- decide when an object is at rest and whether it is speeding up or slowing down.
- interpret a derivative as a rate of change in an application.

Vocabulary · fill in the missing word

Velocity — $v(t) = s'(t)$, the rate of change of _____

Acceleration — $a(t) = v'(t)$, the rate of change of _____

Speed — $|v(t)|$, the _____ value of velocity

Marginal cost — $C'(x)$, the _____ cost of producing one more item

Part 1 · Moving dot

$s(t) = t^3 - 9t^2 + 24t$: watch the dot.

Time interval	Moving right or left?	Sign of v(t)
$0 < t < 2$		
$2 < t < 4$		
$t > 4$		

Part 2 · Notes and examples (fill in each note, then work the example)

Velocity and acceleration: $v(t) = s'(t)$ and $a(t) =$ _____.

Example 1

$s(t) = t^3 - 9t^2 + 24t$ (meters, seconds). Find $v(t)$, $a(t)$, $v(1)$, and $a(1)$.

At rest: the object is at rest when $v(t) =$ _____.

Example 2

For $s(t) = t^3 - 9t^2 + 24t$, when is the object at rest?

Speeding up or slowing down: speeding up when v and a have the _____ sign; slowing down when the signs are opposite.

Example 3

For $s(t) = t^3 - 9t^2 + 24t$, is the object speeding up or slowing down at $t = 1$? At $t = 5$?

Rates of change: the derivative dy/dx is the rate of change of y with respect to _____.

Example 4

Find the rate of change of the area of a circle with respect to its radius when r = 5 cm.

Example 5

$C(x) = 0.02x^2 + 10x + 500$ dollars. Find and interpret $C'(100)$.

Part 3 · Partner practice

Partner 1

$s(t) = t^3 - 6t^2 + 9t$. When is the object at rest?

Partner 2

(AP® Practice) A particle's position is $s(t) = t^3 - 6t^2 + 9t$. At $t = 4$, the particle is

(A) moving left and speeding up (B) moving right and speeding up

(C) moving right and slowing down (D) at rest